# Hyperspectral Image Classification via Neighborhood Adaptive Graph Isomorphism Network

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***Abstract*— The graph convolutional network (GCN) has garnered significant attention in hyperspectral image (HSI) classification due to their ability to model non-Euclidean structured data. Compared with convolutional neural network (CNN), GCN can perform convolutions over irregular image regions and learn global dependencies among pixels in the whole image. Most existing GCN-based methods in the HSI community rely on average aggregation or weighted average aggregation strategies to aggregate neighboring node features. This process tends to obscure the differences between the nodes. However, for HSI classification tasks with obvious intraclass variability, average aggregation is a suboptimal choice. Moreover, the quality of the initial graph structure plays a crucial role in the model's capacity to represent spectral relationships effectively. To mitigate these issues, we propose a neighborhood adaptive graph isomorphism network (NAGIN) for HSI classification to ensure that the diversified spectra representation of land cover can be effectively captured. The neighborhood adaptive block (NAB) enhances spectral discriminability between land-cover classes via spectral reconstruction, enabling more precise removal of anomalous pixels in neighboring nodes. The graph isomorphism network (GIN) aggregates the features of neighboring nodes in an isomorphic manner to obtain multiple spectral expressions of the same type of land cover, ensuring that the spectral features of different land-cover classes can be accurately distinguished. The Kolmogorov–Arnold network (KAN) leverages its ability to learn adaptive activation functions to better extract and refine the spectral features aggregated by GIN. Experimental results demonstrate that NAB can effectively improve the quality of the graph structure, the GIN aggregation method is competitive in HSI classification, and the proposed NAGIN outperforms the state-of-the-art methods on several public HSI datasets.**

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***Index Terms*— Graph convolutional network (GCN), graph isomorphism network (GIN), hyperspectral image (HSI) classification, Kolmogorov–Arnold network (KAN).**

## I. Introduction

HYPERSPECTRAL image (HSI) provides rich spectral information through hundreds of continuous narrow bands, which can effectively distinguish the subtle differences of land cover [1]. Therefore, it has been widely used in urban planning, target detection, and environmental monitoring [2], [3], [4]. However, due to the complexity of spectral information and the diversity of spatial variations, HSI classification tasks face significant challenges.

Traditional HSI classification is mainly based on machine learning classification methods, such as support vector machine (SVM) [5], random forest (RF) [6], and decision tree (DT) [7]. These algorithms capture the nonlinear features in HSI by building complex models, thereby improving the accuracy of classification. However, machine learning classification methods rely on manually designed features and cannot fully explore the deep information in HSI data [8].

Deep learning models can automatically learn hierarchical features from images through multiple layers of nonlinear transformations [9]. Chen et al. [10] first introduced the concept of deep learning into HSI classification by employing stacked autoencoders to extract deep features. However, autoencoders overlook spatial relationships between pixels during processing. In contrast, a convolutional neural network (CNN), with its local receptive field structure, can effectively capture spatial information and has been widely adopted for spectral–spatial feature extraction in HSI. For instance, 2-D CNN is utilized to extract high-level spatially correlated features [11], while 3-D CNN simultaneously models spectral and spatial dimensions for more comprehensive feature learning [12]. While CNN excels in extracting local spectral–spatial features, HSI classification also relies on global spectral and spatial relationships. The transformer architecture addresses this limitation by modeling global context through self-attention mechanisms [13]. Existing methods have designed spectral–spatial feature extraction networks based on the transformer to effectively capture global semantic information, thereby enhancing feature representation [14]. Nevertheless,

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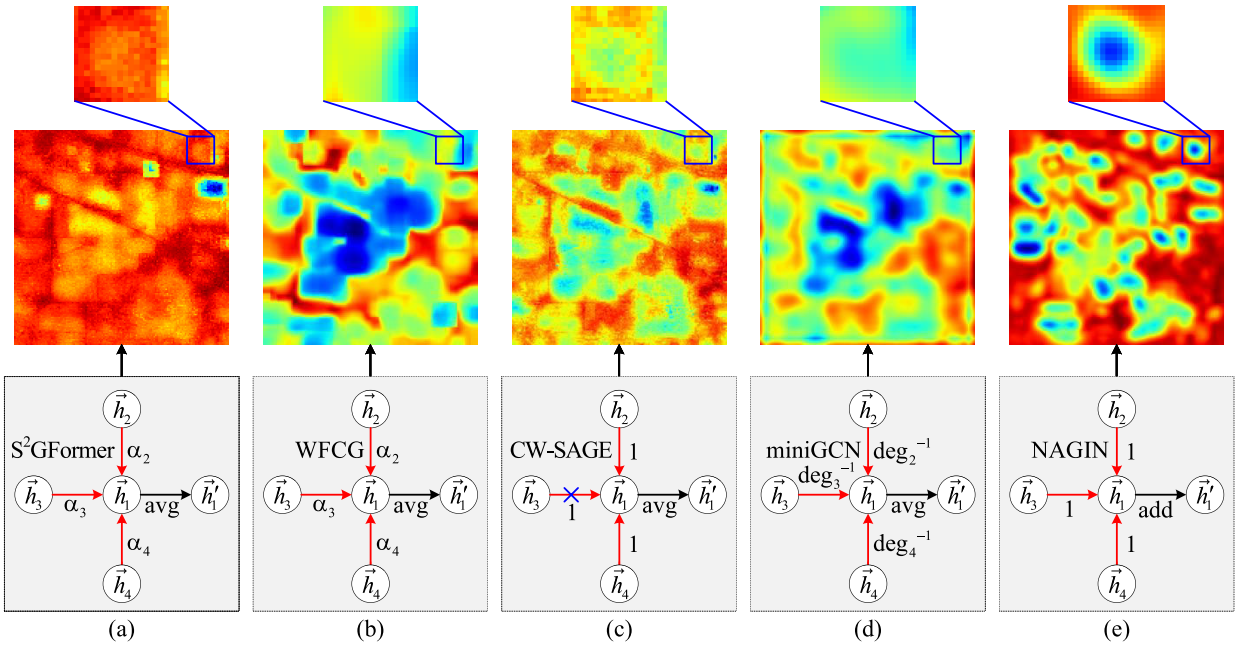



Fig. 1. GCN-based aggregation framework and feature maps, where the feature maps represent heatmaps of node features obtained after two-layer graph convolutional. S²GFormer and WFCG aggregate node features by learning edge weights, CW-SAGE aggregates node features by randomly masking some edges, and miniGCN considers the degree of each node during the aggregation process. Overall, these methods aggregate node features by averaging or weighted averaging, and the features obtained are relatively simple, as illustrated in (a)–(d). NAGIN combines different features by summing aggregation and can obtain diversified feature representations of the same type of land cover, as illustrated in (e). The blue outlined area are the same type of land cover, after multiple layers of graph convolution, NAGIN produces more enriched features.

the transformer typically processes fixed-size patches as input, primarily modeling global relationships among pixels within localized regions, which may still lack sufficient structural perception across the entire image.

Graph convolutional network (GCN) performs convolutional operations on graph-structured data, effectively mining spatial dependencies between arbitrary nodes [15]. In HSI classification, GCN has been employed to construct spectral–spatial graph structures, capturing global spatial patterns through feature aggregation among adjacent nodes [16]. Some approaches build graph models from a superpixel perspective to jointly model local and global features [17], while others combine CNN and GCN to separately extract local features within superpixels and global relationships between them [18]. Despite their strengths in global modeling, the current GCN-based classification model uses average aggregation or weighted average aggregation to update node features. This process tends to obscure the differences between nodes. In particular, when the graph structures are similar but there are different node features, GCN may not be able to effectively distinguish them, which leads to different graph structures being considered isomorphic, limiting the classification performance of the model. Second, anomalous pixels in the initial graph will limit the classification performance of the model. While existing methods have studied graph structure optimization problems, they have not effectively removed the anomalous pixels in the initial graph.

We propose a neighborhood adaptive graph isomorphism network (NAGIN) model, which is an efficient framework designed to capture the diverse spectral representation of land cover. The core structure of NAGIN includes a neighborhood adaptive block (NAB) and a Kolmogorov–Arnold network (KAN)-based graph isomorphism network (GIN) model. Specifically, NAB enhances the discriminative capability between spectral features of different classes by reconstructing spectral features and then improves the quality of the initial graph by removing edges between nodes of different classes based on a spectral similarity metric. The GIN uses a summation aggregation strategy in each layer of graph convolution, which can more effectively integrate and capture the diverse spectral characteristics of similar land cover compared with the commonly used average aggregation or weighted average aggregation methods, as illustrated in Fig. 1. This aggregation approach enables GIN to better model the spectral relationships between both similar and dissimilar land-cover classes, resulting in more accurate node embedding representations.

The main contributions of this work are summarized as follows.

1) A GCN-based HSI classification framework (NAGIN) is proposed, which utilizes spectral difference characteristics, aggregates the global features of nodes in a homogeneous manner, and constructs multiple spectral expressions of the same land-cover classes to improve classification accuracy.
2) A graph-based NAB is proposed to remove anomalous pixels in neighboring nodes by spectral reconstruction, effectively improving the quality of the adjacency matrix.
3) Enhance the feature extraction approach in GIN using KAN by leveraging its capability to learn adaptive activation functions, thereby more effectively capturing nonlinear relationships in HSI.

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4) Experiments on three standard datasets demonstrate that our method effectively improves the quality of the initial graph and enhances the aggregation performance of GCN. In addition, experiments on the 2018 IEEE GRSS data fusion contest dataset show that the proposed method exhibits strong generalization performance.

The remaining sections of this article are structured as follows. Section II reviewed the related work. Section III provides a comprehensive description of the proposed method. Section IV outlines the experimental results and analyses. Conclusion and future work are discussed in Section V.

## II. Related Works

### A. HSI Classification Based on CNN

The rise of deep learning, especially the widespread application of CNN, makes it possible to automatically extract spatial and spectral features [19]. CNN extracts local features through convolution kernels and performs feature fusion at a deep level, which can well adapt to the high-dimensional spectral properties of HSI [20], [21], [22]. To alleviate the overfitting and gradient vanishing problems of deep learning, Paoletti et al. [23] introduced shortcut connections between layers, taking the feature map of the next layer as the input of the previous layer, combining different spectral–spatial features across layers, and enhancing the generalization ability of the model. In response to the imbalance problem between high dimensionality and limited training samples commonly seen in HSI classification, a regularized 3-D CNN model is proposed to effectively extract the spectral–spatial features of HSI [24], [25]. To further optimize the cross-channel dependency of the model, Roy et al. [26] proposed an improved residual network based on an adaptive spectral–spatial kernel based on attention, which captured discriminative spectral–spatial features in an end-to-end training manner and adopted a feature calibration mechanism to improve the classification performance. To enhance the training efficiency and mitigate the negative impact of network depth on classification accuracy, subsequent studies proposed an end-to-end fast dense spectral–spatial convolutional framework. This framework extracts spectral and spatial features separately using convolutional kernels of varying scales, thereby optimizing computational efficiency [27]. Further improved models adopted a more refined spectral–spatial feature extraction strategy. Based on residual structures, they progressively expanded the feature dimensions of convolutional layers and introduced pyramid bottleneck residual block grouping, thereby maintaining reasonable time complexity while increasing network depth [28]. However, CNN mainly extracts local features based on convolution kernels, and its global feature extraction ability is not sufficient.

### B. HSI Classification Based on Transformer

Transformer captures global spectral–spatial features of HSI through self-attention mechanisms [29]. Roy et al. [30] proposed a learnable spectral–spatial morphological network based on the transformer, which employs spectral and spatial morphological convolutions to enhance HSI classification performance. To fully exploit the spatial and spectral characteristics of HSI, a spatial–spectral transformer classification framework was introduced, utilizing a densely connected transformer to capture sequential spectral relationships while incorporating dynamic enhancement techniques to mitigate overfitting issues [31]. For the collaborative extraction of hybrid multiscale spatial and spectral features, a hybrid multi–scale spatial–spectral framework was developed, innovatively modeling global dependencies at multiple scales from both spatial and spectral domains [32]. To address the high computational complexity of transformer, Zhang et al. [33] proposed two types of lightweight self-attention modules, channelwise multihead self-attention, and positionwise lightweight multihead self-attention, to reduce computational overhead while acquiring global spectral and spatial relationships. In response to the potential over-dispersion of features extracted by transformer multihead attention, a group-aware hierarchical transformer for HSI classification was introduced, confining multihead self-attention to local spatialspectral contexts through a novel grouped pixel embedding module [34]. However, CNN and transformer models that use patches as input can only extract local spatial features and cannot truly model global spatial relationships.

### C. HSI Classification Based on GCN

GCN effectively captures spatial dependencies between nodes by taking non-Euclidean graph-structured data as input [15]. At present, a variety of HSI classification methods based on GCN have been proposed [35], [36], [37], [38], [39]. Specifically, Ding et al. [40] proposed a graph convolutional smoothing method for class imbalance, leveraging the global feature extraction capability of GCN to achieve adaptive feature smoothing in imbalanced class regions. Hong et al. [41] developed a mini-GCN model that enables mini-batch training, thereby improving training efficiency. To address the impact of graph structure quality on classification, a diversity-connected GCN model was introduced. This model explores potential neighbors through topological expansion to optimize the graph structure [42]. In addition, a spatial–spectral adaptive graph convolutional subspace clustering method was proposed, which dynamically updates the graph convolutional operator through the reconstruction coefficient matrix, enabling the joint optimization of feature representations and the coefficient matrix [43]. Some studies have also adopted metaheuristic optimization strategies to adjust the adjacency matrix, thereby constructing an adaptive GCN framework [44]. In the field of few-shot classification, Shi et al. [45] designed a multihop graph attention and spectral-overlapping grouped convolutional fusion network for HSI classification. This network uses multihop graph attention to weight and refine the features extracted by graph convolution and builds a spectral intragroup and intergroup feature extraction module based on spectral overlapping groups. In response to the problem that the current GCN node feature aggregation can only extract shallow features, Chen et al. [46] proposed a new topology-feature-aware pooling graph U-Net, which considers the topological

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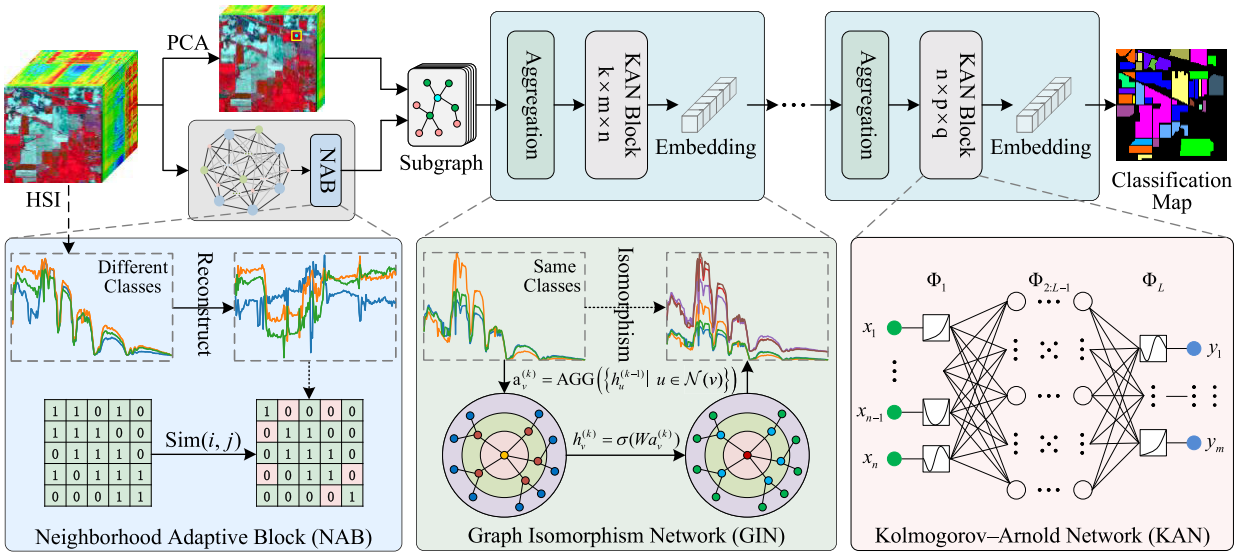



Fig. 2. Overall architecture of the proposed NAGIN framework for HSI classification. NAB enhances interclass spectral variability by reconstructing spectral features, thereby refining the adjacency matrix. GIN employs the isomorphic learning to capture diverse representations of spectra within the same class, while KAN improves feature extraction from GIN-aggregated representations through learnable activation functions.

structure of the graph and the node feature information to automatically select nodes that are beneficial to classification.

## III. Proposed Network

This section first introduces the representative working method of GCN, then introduces the overall architecture of this article, and finally introduces the research content of this article from three aspects: graph structure optimization, information aggregation, and feature extraction.

### A. Preliminaries

GNN performs graph convolutional operations by aggregating features from nodes and their neighbors. By stacking multiple graph convolutional layers, GNN captures higher order relationships between nodes in the graph, thereby learning deeper graph features. GNN has demonstrated excellent performance in tasks, such as node classification, graph classification, and link prediction. The working principle of GCN is.

Given an undirected graph $G = (V, E)$, where $V$ and $E$ represent nodes and edges, respectively. In this article, each node represents a pixel in the HSI, and each edge reflects the similarity between two nodes $v_i$ and $v_j$. The corresponding adjacency matrix $A$ is a sparse matrix of size $N \times N$, where $N$ is the number of nodes. If there is an edge between nodes $v_i$ and $v_j$, then $A_{ij} = 1$; otherwise, $A_{ij} = 0$. Each node is associated with a $d$-dimensional feature vector and $X \in \mathbb{R}^{N\times d}$ represents the initial feature matrix of all $N$ nodes. GCN consists of $K$ graph convolutional layers, where each layer builds new node embeddings by aggregating the features of the neighboring nodes in the previous layer. Specifically, the output of the $k$th layer can be represented as follows:

$$H^{(k)} = \sigma\left(\tilde{A}H^{(k-1)}W^{(k)}\right) \tag{1}$$

where $H^{(k)}$ represents the embedding of all $N$ nodes in the $k$th layer, $H^{(0)} = X$, $W^{(k)}$ is the learnable weight matrix, $\sigma$ is an activation function, and $\tilde{A}$ is the normalized adjacency matrix, which can be expressed as follows:

$$\tilde{A} = D^{-\frac{1}{2}}(A + I_N)D^{-\frac{1}{2}} \tag{2}$$

where $I_N$ is the identity matrix and $D$ is the degree matrix of $A + I_N$.

### B. Proposed NAGIN

Although GCN-based methods can extract global and local features from non-Euclidean graph-structured data, the approaches for node aggregation and feature extraction remain worthy of exploration. Therefore, we investigate novel node aggregation strategies and propose a pixel-level NAGIN model for HSI classification, which consists of three modules: NAB, GIN, and KAN. The NAB is designed to eliminate anomalous pixels among neighboring nodes, while the GIN aggregates node information in an isomorphic manner to obtain diversified node representations. The KAN is employed to efficiently extract features from the aggregated nodes in GIN. The overall framework of the model is illustrated in Fig. 2, and the structure of the KAN-based GIN network is depicted in Fig. 3.

### C. Neighborhood Adaptive Block

Traditional deep learning models are based on Euclidean structured data modeling, and their input must be data of the same size. In the HSI classification task, based on the spatial homogeneity assumption, patches of the same size are used as input. This approach may mix anomalous pixels in the patches, limiting the classification performance of the model. GCN is based on non-Euclidean graph-structured data modeling and uses a graph that represents relationships between nodes as input, which allows us to flexibly construct the adjacency relationship of each node and avoid the interference of anomalous

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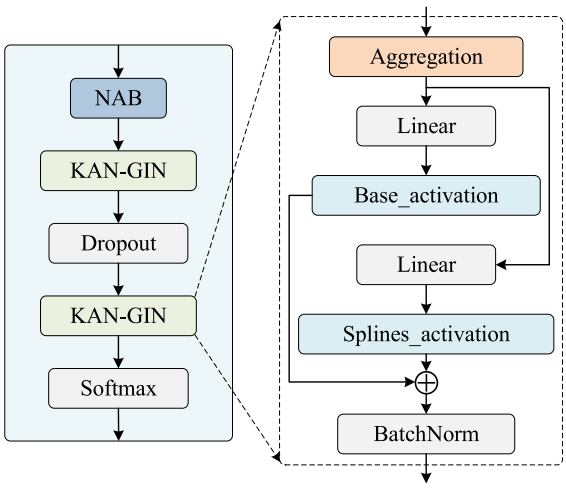



Fig. 3. Structure of the KAN-based GIN network.

pixels. Specifically, the NAB first reconstructs the spectrum to improve the distinction between spectra and then uses the metric function to remove anomalous pixels in neighboring nodes to improve classification accuracy. Fig. 4(a) shows a spectral curve obtained by taking a pixel from each of the 16 different land-cover classes in the Indian Pines dataset. It can be seen that the spectral curves of the 16 land-cover classes are very similar. To more accurately distinguish the spectra of different land-cover classes, we employed four existing feature scaling methods along with the proposed NAB to reconstruct the spectral features, as shown in Fig. 4(b)–(f). It can be seen that the reconstructed spectral features are easier to distinguish. The processing steps of the NAB are as follows.

Assume that an HSI contains $H \times W$ pixels, where $H$ and $W$ represent the height and width, respectively. Moreover, each pixel $x_i$ is a $B$-dimensional spectral vector. For each pixel $x_i$, the mean $\mu_b$ and standard deviation $\sigma_b$ for each band $b$ are given by

$$\mu_b = \frac{1}{B}\sum_{b=1}^{B} x_{i,b} \tag{3}$$

where $x_{i,b}$ is the value of the $b$th band of the $i$th pixel, and the standard deviation $\sigma_b$ is given by

$$\sigma_b = \sqrt{\frac{1}{B}\sum_{b=1}^{B}\left(x_{i,b} - \mu_b\right)^2}. \tag{4}$$

Each band is then reconstructed as follows:

$$x'_{i,b} = \frac{x_{i,b} - \mu_b}{\sigma_b} \tag{5}$$

where $x_{i,b}$ represents the value of the $i$th pixel at band $b$ before feature reconstruction and $x'_{i,b}$ is the value after reconstruction.

Subsequently, the spectral metric method is used to calculate the similarity between each node and its neighbors. For any two pixels $x_i$ and $x_j$, the cosine similarity between the reconstructed $x'_i$ and $x'_j$ is given by

$$\mathrm{Sim}\left(x'_i, x'_j\right) = \frac{x'_i \cdot x'_j}{\|x'_i\|\,\|x'_j\|}. \tag{6}$$

The NAB method reconstructs the spectral features of each pixel into a normal distribution with a mean of 0. Thus, pixels with a threshold less than 0 are considered anomalous and removed from the neighborjing nodes to optimize the adjacency matrix, thereby enhancing the model's classification performance, which can be formulated as follows:

$$A[i][j] = \begin{cases} 0, & \text{if } \cos\left(\theta_{ij}\right) < 0 \\ A[i][j], & \text{otherwise.} \end{cases} \tag{7}$$

### D. Graph Isomorphism Network

HSI generally has the phenomenon of different spectra of the same land-cover classes and different land-cover classes of the same spectra. Accurately distinguishing different types of land cover is a challenging problem. Ideally, the most powerful GCN can map different graph structures to different embedding spaces. However, the ability to map any two different graphs to different embedding spaces means solving the challenging graph isomorphism problem. By utilizing the Weisfeiler–Lehman (WL) graph isomorphism algorithm to assess the model's representational power, we designed a GIN specifically for HSI classification. GIN learns diversified embedding representations of input features by combining the adjacency relationships of different nodes, as shown in Fig. 5. This process can be described as follows.

Each pixel of the HSI is treated as a node, with its features denoted by $x_i$. The GCN aggregates information from neighboring nodes using an aggregation strategy and extracts spectral features through a neural network. After $k$ iterations of aggregation, GCN can capture the structural information within its $k$-hop neighborhood. The representation of the $k$th layer of the GCN is as follows:

$$a_v^{(k)} = \mathrm{AGG}\left(\left\{h_u^{(k-1)} | u \in \mathcal{N}(v)\right\}\right) \tag{8}$$

$$h_v^{(k)} = \sigma\left(W a_v^{(k)}\right) \tag{9}$$

where $h_v^{(k)}$ is the feature vector of node $v$ at the $k$th layer. We initialize $h_v^{(0)} = x_v$, $W$ is the weight matrix, $\sigma$ is the activation function, and $\mathcal{N}(v)$ is the set of nodes adjacent to $v$.

For HSI classification tasks, based on the spatial homogeneity assumption, we consider any node and its one-hop neighbors as belonging to the same type of land cover. For HSI with significant intraclass differences, the choice of aggregation function in GCN is crucial. To further explore the expressive power of GCN, we analyze the relationship between GCN and the WL test. Suppose there are two nonisomorphic graphs $G_1$ and $G_2$. If a GCN can map these two graphs to different embedding spaces, i.e., $h(G_1) \neq h(G_2)$, it implies that the node feature sets of these two graphs must differ at some layer. The WL test updates node labels and assigns unique new labels at each iteration, whereas the GCN updates each node feature by aggregating the feature vectors of its neighboring nodes. If a GCN can distinguish between $G_1$ and $G_2$, then at some level, the sets of node features in the two graphs must differ, which can be formulated as follows:

$$\left\{h_v^{(k)} | v \in G_1\right\} \neq \left\{h_u^{(k)} | u \in G_2\right\}. \tag{10}$$

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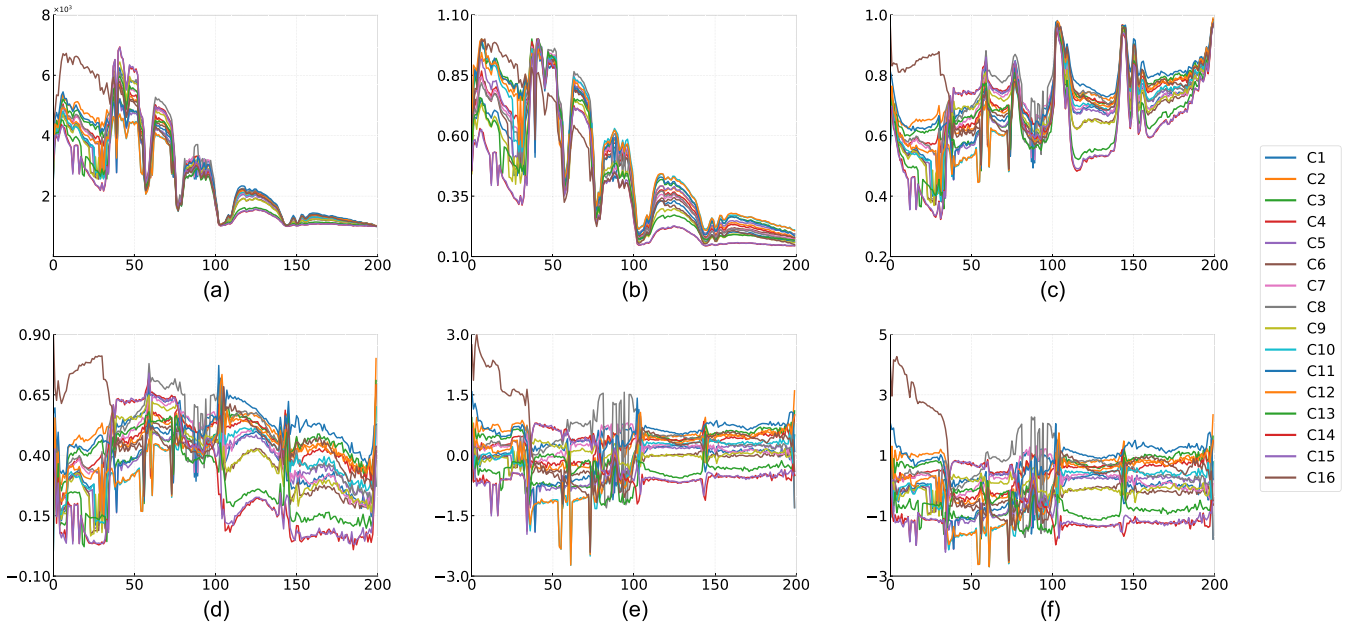



Fig. 4. Spectral reflectance reconstructed by different methods. (a) Original spectral curve. (b) Normalize. (c) Maxabs. (d) Minmax. (e) Robust. (f) NAB.

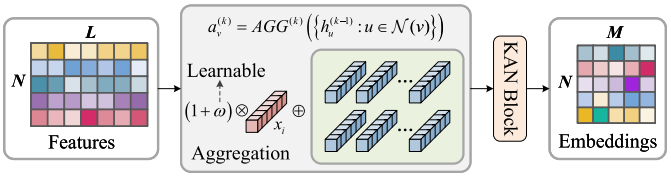



Fig. 5. GIN aggregation method.

According to the properties of the WL test, if the node label sets differ at some iteration, the WL test can distinguish between the two graphs. Therefore, if a GCN can distinguish between $G_1$ and $G_2$, the WL test can also distinguish them. Hence, any GCN based on aggregation has, at most, the discriminative power of the WL test.

Next, we discuss what kind of GCN can achieve the same discriminative power as the WL test. Suppose a GCN has an injective aggregation function. In this case, different sets of node neighbors will generate different feature vectors at each aggregation step, similar to the label updates in the WL test. The aggregation process of the GCN updates node features based on the features of neighboring nodes. For any two nonisomorphic graphs $G_1$ and $G_2$, as long as the WL test can distinguish them, the injective aggregation process of the GCN will also distinguish them. Thus, a GCN satisfying the injective condition can attain maximum expressive capacity, equivalent to the discriminative power of the WL test.

What kind of aggregation function is injective? We analyze the summation aggregation function. Suppose the summation aggregation function relies on a constructor $f$, which maps each element in a multiset to a real vector. To ensure uniqueness, $f$ is defined as follows:

$$f(a_i) = \sum_{k=1}^{d} w_k \cdot a_i^k \tag{11}$$

where $w_k$ is a large positive integer and $a_i^k$ is the natural number identifier of element $a_i$. This ensures that different sets of elements will generate different sums.

Even if there are identical elements in the multiset, the sum will produce different results based on the frequency of occurrence and specific values, which can be formulated as follows:

$$h_{\mathcal{N}(v)} \neq h_{\mathcal{N}(u)} \tag{12}$$

where $h_{\mathcal{N}(v)}$ and $h_{\mathcal{N}(u)}$ represent different sets of neighboring nodes, thereby enabling the construction of an aggregation function that fulfills the injective condition.

### E. Kolmogorov–Arnold Network

HSI contains rich spectral information, and the ability to extract spectral features is crucial for the classification performance of the model. KAN learns variable activation functions to more effectively capture complex relationships between spectral bands. After aggregating spectral information, the GCN needs to effectively map it to the next layer of nodes. We use KAN to extract aggregated features. KAN is based on the Kolmogorov–Arnold representation theorem, which states that any multivariate continuous function can be expressed as a finite composition of univariate continuous functions. The KAN leverages this property by applying learnable activation functions to the edges, which can be formulated as follows:

$$h_b(x) = \sum_{i=1}^{k} \alpha_i f_i(x) \tag{13}$$

where $f_i$ is a univariate function representing the output of the $i$th spectral band feature after applying the univariate function. The function $h_b(x)$ represents the result of combining the outputs from multiple bands.

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In a traditional neural network, activation functions are fixed, but in the KAN model, activation functions are learnable. This means that each function $f_i(x)$ is a parameterized function optimized during the training process, which can be formulated as follows:

$$f_i(x) = \text{Spline}(x) \tag{14}$$

where Spline represents a spline function, expressed through B-spline basis functions, and its parameters are learned during training. The spline function can be formulated as follows:

$$\text{Spline}(x) = \sum_{j=1}^{m} c_j B_j(x) \tag{15}$$

where $B_j(x)$ is the $j$th B-spline basis function and $c_j$ is the coefficients learned during training.

The training process of the KAN model involves minimizing a loss function. The mean squared error is commonly used as the loss function, which can be formulated as follows:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} (h_b(x_i) - y_i)^2 \tag{16}$$

where $h_b(x_i)$ is the output of KAN and $y_i$ is the true label. Through the backpropagation algorithm, the parameters of the model are adjusted so that the model can accurately fit the spectral features.

## IV. Performance Evaluation

### A. Datasets Description

To evaluate the classification performance of the proposed NAGIN, we evaluate the performance of NAGIN on three standard datasets (Indian Pines, WHU-Hi-HongHu, and WHU-Hi-HanChuan) and one competition dataset (Houston 2018). For each dataset, 1% of the samples are used for training, while the remaining samples are used for testing.

*1) Indian Pines:* This dataset is the first test data used for HSI classification. It is an Indian pine dataset obtained by annotating 145 × 145 images captured by the AVIRIS sensor in the agricultural area of northwest Indiana in 1992. The wavelength range is 0.4–2.5 $\mu$m, the spatial resolution is 20 m, and it contains a total of 220 bands. After removing 20 water absorption bands, 200 bands are retained, with a labeling ratio of 48.75% and 16 types of land cover. The number of training and testing samples for each class in the Indian Pines dataset is shown in Table I.

*2) WHU-Hi-HongHu:* The WHU-Hi-HongHu dataset was acquired from 16:23 to 17:37 on November 20, 2017, in Honghu City, Hubei province, China, with a 17-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro UAV platform [47]. The experimental area is a complex agricultural scene with many classes of crops, and different cultivars of the same crop are also planted in the region. The UAV flew at an altitude of 100 m, the size of the imagery is 940 × 475 pixels, there are 270 bands from 400 to 1000 nm, and the spatial resolution of the UAV-borne HSI is about 0.043 m. The labeling ratio is 86.61%, including 22 types of land-cover classes. The number of training and testing samples for each class in the WHU-Hi-HongHu dataset is shown in Table II.

TABLE I
Class Name, Sample Numbers of Training Set, and Testing Set of Each Class on the Indian Pines Dataset

| Class No. | Class Name | Train | Test | Total |
|---|---|---|---|---|
| 1 | Alfalfa | 1 | 45 | 46 |
| 2 | Corn-notill | 15 | 1413 | 1428 |
| 3 | Corn-mintill | 9 | 821 | 830 |
| 4 | Corn | 3 | 234 | 237 |
| 5 | Grass-pasture | 5 | 478 | 483 |
| 6 | Grass-trees | 8 | 722 | 730 |
| 7 | Grass-pasture-mowed | 1 | 27 | 28 |
| 8 | Hay-windrowed | 5 | 473 | 478 |
| 9 | Oats | 1 | 19 | 20 |
| 10 | Soybean-notill | 10 | 962 | 972 |
| 11 | Soybean-mintill | 25 | 2430 | 2455 |
| 12 | Soybean-clean | 6 | 587 | 593 |
| 13 | Wheat | 3 | 202 | 205 |
| 14 | Woods | 13 | 1252 | 1265 |
| 15 | Building-Grass-Trees-Drives | 4 | 382 | 386 |
| 16 | Stone-Steel-Towers | 1 | 92 | 93 |
| Total | | 110 | 10139 | 10249 |

TABLE II
Class Name, Sample Numbers of Training Set, and Testing Set of Each Class on the WHU-Hi-HongHu Dataset

| Class No. | Class Name | Train | Test | Total |
|---|---|---|---|---|
| 1 | Red roof | 141 | 13900 | 14041 |
| 2 | Road | 36 | 3476 | 3512 |
| 3 | Bare soil | 219 | 21602 | 21821 |
| 4 | Cotton | 1633 | 161652 | 163285 |
| 5 | Cotton firewood | 63 | 61552 | 6218 |
| 6 | Rape | 446 | 44111 | 44557 |
| 7 | Chinese cabbage | 242 | 23861 | 24103 |
| 8 | Pakchoi | 41 | 4013 | 4054 |
| 9 | Cabbage | 109 | 10710 | 10819 |
| 10 | Tuber mustard | 124 | 12270 | 12394 |
| 11 | Brassica parachinensis | 111 | 10904 | 11015 |
| 12 | Brassica chinensis | 90 | 8864 | 8954 |
| 13 | Small Brassica chinensis | 226 | 22281 | 22507 |
| 14 | Lactuca sativa | 74 | 7282 | 7356 |
| 15 | Celtuce | 11 | 991 | 1002 |
| 16 | Film covered lettuce | 73 | 7189 | 7262 |
| 17 | Romaine lettuce | 31 | 2979 | 3010 |
| 18 | Carrot | 33 | 3184 | 3217 |
| 19 | White radish | 88 | 8624 | 8712 |
| 20 | Garlic sprout | 35 | 3451 | 3486 |
| 21 | Broad bean | 14 | 1314 | 1328 |
| 22 | Tree | 41 | 3999 | 4040 |
| Total | | 3881 | 382812 | 386693 |

*3) WHU-Hi-HanChuan:* The WHU-Hi-HanChuan dataset was acquired from 17:57 to 18:46 on June 17, 2016, in Hanchuan, Hubei province, China, with a 17-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a Leica Aibot X6 UAV V1 platform. The study area is a rural–urban fringe zone with buildings, water, and cultivated land. The UAV flew at an altitude of 250 m, the size of the imagery is 1217 × 303 pixels, there are 274 bands from 400 to 1000 nm, and the spatial resolution of the UAV-borne HSI is about 0.109 m. The labeling ratio is 69.84%, including 22 types of land-cover classes. The number of training and testing samples for each class in the WHU-Hi-HanChuan dataset is shown in Table III.

*4) Houston 2018:* This dataset was provided by the Image Analysis and Data Fusion Technical Committee in the 2018 IEEE GRSS data fusion contest, which are the images of the University of Houston Energy Research Park (UHEP)

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TABLE III
CLASS NAME, SAMPLE NUMBERS OF TRAINING SET, AND TESTING SET OF EACH CLASS ON THE WHU-HI-HANCHUAN DATASET

| Class No. | Class Name | Train | Test | Total |
|---|---|---|---|---|
| 1 | Strawberry | 448 | 44287 | 44735 |
| 2 | Cowpea | 228 | 22525 | 22753 |
| 3 | Soybean | 103 | 10184 | 10287 |
| 4 | Sorghum | 54 | 5299 | 5353 |
| 5 | Water spinach | 12 | 1188 | 1200 |
| 6 | Watermelon | 46 | 4487 | 4533 |
| 7 | Greens | 60 | 5843 | 5903 |
| 8 | Trees | 180 | 17798 | 17978 |
| 9 | Grass | 95 | 9374 | 9469 |
| 10 | Red roof | 106 | 10410 | 10516 |
| 11 | Gray roof | 170 | 16741 | 16911 |
| 12 | Plastic | 37 | 3642 | 3679 |
| 13 | Bare soil | 92 | 9024 | 9116 |
| 14 | Road | 186 | 18374 | 18560 |
| 15 | Bright object | 12 | 1124 | 1136 |
| 16 | Water | 755 | 74646 | 75401 |
| Total | | 2854 | 254946 | 257530 |

and the Earth and Atmospheric Science building (UH01) and one temporary station located at the Baytown airport (KHPY) [48]. The size of the imagery is 4172 × 1202 pixels, there are 48 bands from 380 to 1050 nm, and the spatial resolution is 1 m. This dataset is only marked in a fixed area of 601 × 2384 pixels, and the marking ratio is 10.07%. The number of training and testing samples for each class in the Houston 2018 dataset is shown in Table IV.

### B. Experimental Settings

*1) Evaluation Metrics:* To verify the performance of NAGIN, we selected three evaluation criteria, including overall classification accuracy (OA), average classification accuracy (AA), and kappa coefficient ($\mathcal{K}$). Among them, OA is the ratio of the number of correctly predicted pixels to the total pixels, AA is the average classification accuracy of all categories, and $\mathcal{K}$ is used to measure the consistency of the classification results with the ground truth. The higher their values, the better the classification performance. All experiments are the average values of ten consecutive experiments with the same proportion of randomly selected training samples under the same experimental conditions.

*2) Implementation Details:* NAGIN and the comparison experiments are run on a server equipped with an NVIDIA A800 (80 GB) GPU using the PyTorch framework. For the GCN classification model, we use the PyTorch Geometric library. For parameters, the Adam optimizer is used, the learning rate is set to 0.005, and the epoch is set to 1000.

*3) Comparison Method:* To demonstrate the effectiveness of NAGIN, we selected four representative HSI classification frameworks for comprehensive comparison, including traditional classification method (SVM) [5], CNN-based classification methods (2-D CNN [21], 3-D CNN [24], and SSRN [12]), transformer-based classification method (SSFTT [14]), and GCN-based classification methods ($S^2$GFormer [49], WFCG [50], CW-SAGE [51], and miniGCN [41]).

TABLE IV
CLASS NAME, SAMPLE NUMBERS OF TRAINING SET, AND TESTING SET OF EACH CLASS ON THE HOUSTON 2018 DATASET

| Class No. | Class Name | Train | Test | Total |
|---|---|---|---|---|
| 1 | Healthy grass | 98 | 9701 | 9799 |
| 2 | Stressed grass | 326 | 32176 | 32502 |
| 3 | Artificial turf | 7 | 677 | 684 |
| 4 | Evergreen trees | 136 | 13459 | 13595 |
| 5 | Deciduous trees | 51 | 4970 | 5021 |
| 6 | Bare earth | 46 | 4470 | 4516 |
| 7 | Water | 3 | 263 | 266 |
| 8 | Residential buildings | 398 | 39374 | 39772 |
| 9 | Non-residential buildings | 2238 | 221514 | 223752 |
| 10 | Roads | 459 | 45407 | 45866 |
| 11 | Sidewalks | 341 | 33688 | 34029 |
| 12 | Crosswalks | 16 | 1502 | 1518 |
| 13 | Major thoroughfares | 464 | 45884 | 46348 |
| 14 | Highways | 99 | 9766 | 9865 |
| 15 | Railways | 70 | 6867 | 6937 |
| 16 | Paved parking lots | 115 | 11385 | 11500 |
| 17 | Unpaved parking lots | 2 | 144 | 146 |
| 18 | Cars | 66 | 6481 | 6547 |
| 19 | Trains | 54 | 5315 | 5369 |
| 20 | Stadium seats | 69 | 6755 | 6824 |
| Total | | 5058 | 499798 | 504856 |

### C. Comparison With Competing Methods

The quantitative accuracies (OA, AA, and $\mathcal{K}$) on the Indian Pines, WHU-Hi-HongHu, and WHU-Hi-HanChuan datasets are summarized in Tables V–VII, with the best results highlighted in bold. Corresponding visualization maps are provided in Figs. 6–8.

*1) Results on Indian Pines:* The classification accuracies of different methods on the Indian Pines dataset are listed in Table V. It can be seen that the OA, AA, and $\mathcal{K}$ of the proposed method are higher than those of the nine comparison methods. Among the 16 land-cover classes, the classification accuracy of six land-cover classes is higher than that of other methods, among which the classification accuracy of Alfalfa and Soybean-notill reaches 100%, and the classification accuracy of wheat is 99.5%. Other categories also achieved satisfactory accuracy. The OA of SVM is 59.23%. The best performance of the three CNN-based classification models (2-D CNN, 3-D CNN, and SSRN) is SSRN, with an OA of 86.16%. The best performance of the four GCN-based methods ($S^2$GFormer, WFCG, CW-SAGE, and miniGCN) is miniGCN, with an OA of 88.85%. The OA of SSFTT is 85.92%, and the NAGIN we proposed is 91.07%.

The classification maps of different methods are shown in Fig. 6. For the convenience of comparison, we show the false color image of Indian Pines and the ground-truth map. It can be seen that the SVM classification results are the worst, and obvious salt-and-pepper noise can be seen. The other methods all use spatial features and have better noise control. The CNN-based and transformer-based classification methods will lose detailed information when using spatial information for classification, as shown in the area in the white outlined area in Fig. 6. The classification results at the boundary of the land cover is also relatively poor, and the outline information of the land cover cannot be seen clearly. The GCN-based classification method can see the detailed information and the outline of the land cover, but the classification results for

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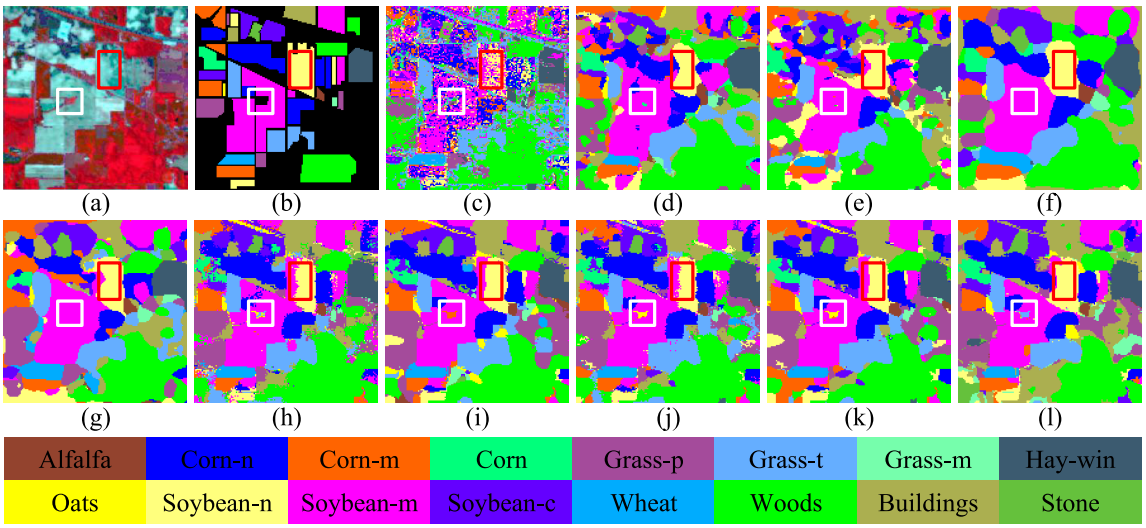



Fig. 6. Classification maps of different methods for the Indian Pines dataset. (a) False-color image. (b) Ground truth. (c) SVM (OA = 59.23%). (d) 2-D CNN (OA = 78.53%). (e) 3-D-CNN (OA = 75.75%). (f) SSRN (OA = 86.16%). (g) SSFTT (OA = 85.92%). (h) S$^2$GFormer (OA = 85.13%). (i) WFCG (OA = 88.76%). (j) CW-SAGE (OA = 88.25%). (k) miniGCN (OA = 88.85%). (l) NAGIN (OA = 91.07%).

TABLE V

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, $\mathcal{K}$, AND THE ACCURACIES OF EACH CLASS FOR THE INDIAN PINES DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

| Class | SVM | 2D-CNN | 3D-CNN | SSRN | SSFTT | S$^2$GFormer | WFCG | CW-SAGE | miniGCN | NAGIN |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 33.26 | 78.48 | 64.57 | 88.70 | 96.30 | 88.89 | 66.67 | 66.67 | 86.67 | **100.0** |
| 2 | 46.09 | 59.08 | 60.36 | 72.05 | 69.19 | 94.20 | 92.14 | 94.90 | 94.76 | **95.54** |
| 3 | 35.40 | 75.22 | 66.04 | 83.86 | 86.17 | 75.76 | 65.41 | 68.21 | 76.00 | **92.20** |
| 4 | 15.65 | 69.41 | 40.13 | 69.24 | 63.38 | 85.47 | **87.61** | 85.04 | 87.18 | 69.66 |
| 5 | 69.25 | 61.82 | 72.22 | 77.41 | 80.08 | 92.47 | **98.54** | 96.65 | 87.87 | 84.94 |
| 6 | 80.66 | 88.47 | 90.47 | 87.71 | 94.84 | 92.52 | 93.63 | **99.17** | 97.78 | 96.12 |
| 7 | 45.71 | 85.00 | 70.36 | 40.71 | 94.64 | 96.30 | **100.0** | **100.0** | **100.0** | 96.30 |
| 8 | 77.41 | 97.68 | 98.83 | 98.60 | 96.09 | **100.0** | **100.0** | **100.0** | **100.0** | 90.70 |
| 9 | 30.50 | 92.00 | 66.00 | 39.00 | 82.00 | 89.47 | 89.47 | 78.95 | 89.47 | **100.0** |
| 10 | 44.15 | 63.42 | 69.53 | 77.98 | 77.83 | 58.42 | 80.35 | 74.01 | 76.51 | **83.16** |
| 11 | 70.10 | 87.76 | 77.99 | 94.29 | 92.54 | 92.22 | **97.41** | 93.66 | 91.77 | 95.76 |
| 12 | 27.88 | 57.66 | 55.90 | **81.11** | 70.61 | 55.54 | 60.48 | 64.40 | 66.61 | 66.44 |
| 13 | 91.17 | 81.56 | 79.12 | 86.00 | 98.68 | 96.53 | 95.05 | 99.01 | 98.51 | **99.50** |
| 14 | 85.94 | 97.15 | 95.93 | 99.42 | 98.58 | 99.28 | **100.0** | 99.60 | 99.84 | 98.48 |
| 15 | 25.10 | 89.30 | 89.17 | 86.48 | **92.59** | 63.87 | 64.14 | 69.11 | 73.56 | 86.13 |
| 16 | 73.12 | 78.49 | 77.53 | 79.78 | 79.03 | 71.09 | **83.70** | 52.17 | 78.26 | 70.65 |
| OA(%) | 59.23 | 78.53 | 75.75 | 86.16 | 85.92 | 85.13 | 88.76 | 88.25 | 88.85 | **91.07** |
| AA(%) | 53.21 | 78.91 | 73.38 | 78.90 | 85.78 | 84.50 | 85.91 | 83.85 | 87.80 | **89.10** |
| $\mathcal{K} \times 100$ | 53.03 | 75.40 | 72.32 | 84.15 | 83.93 | 82.89 | 87.09 | 86.49 | 87.21 | **89.82** |

the large area are not particularly ideal, as shown in the red outlined area in Fig. 6. The best classification results in the red outlined area are SSRN, but SSRN can hardly see the true outline of the land cover. Overall, NAGIN demonstrates the best classification performance, maintaining both precise boundary delineation with detailed feature preservation while achieving high-classification accuracy.

*2) Results on WHU-Hi-HongHu:* The classification accuracy of different methods on the WHU-Hi-HongHu dataset is listed in Table VI. It can be seen that the OA, AA, and $\mathcal{K}$ of this method are higher than the nine comparison methods. Among the 22 land-cover classes, the classification accuracy of 19 land-cover classes exceeds that of other methods. Only the classification accuracies of road, cotton, and film covered lettuce are slightly different from the best classification accuracy of the comparison methods. The OA of SVM is 84.02%, the best of the three CNN-based classification models is SSRN, with an OA of 96.08%, the best of the four GCN-based classification methods is WFCG, with an OA of 95.90%, and the OA of SSFTT is 94.57%. The OA of the NAGIN classification model we proposed is 98.34%.

The classification results of different methods are shown in Fig. 7. For the convenience of comparison, we show the WHU-Hi-HongHu false color image and ground-truth map. It can be seen that SVM has the worst classification results, and an obvious salt-and-pepper noise can be seen. There is no obvious difference between the three CNN-based classification methods and SSFTT. Among the four GCN-based classification models, S$^2$GFormer has the worst classification results, with obvious noise and misclassification. The test accuracy of WFCG is the highest among the four GCN-based classification models, but the classification results are not the best. The test accuracy of CW-SAGE is lower than that of WFCG and miniGCN, but the classification results are better than those

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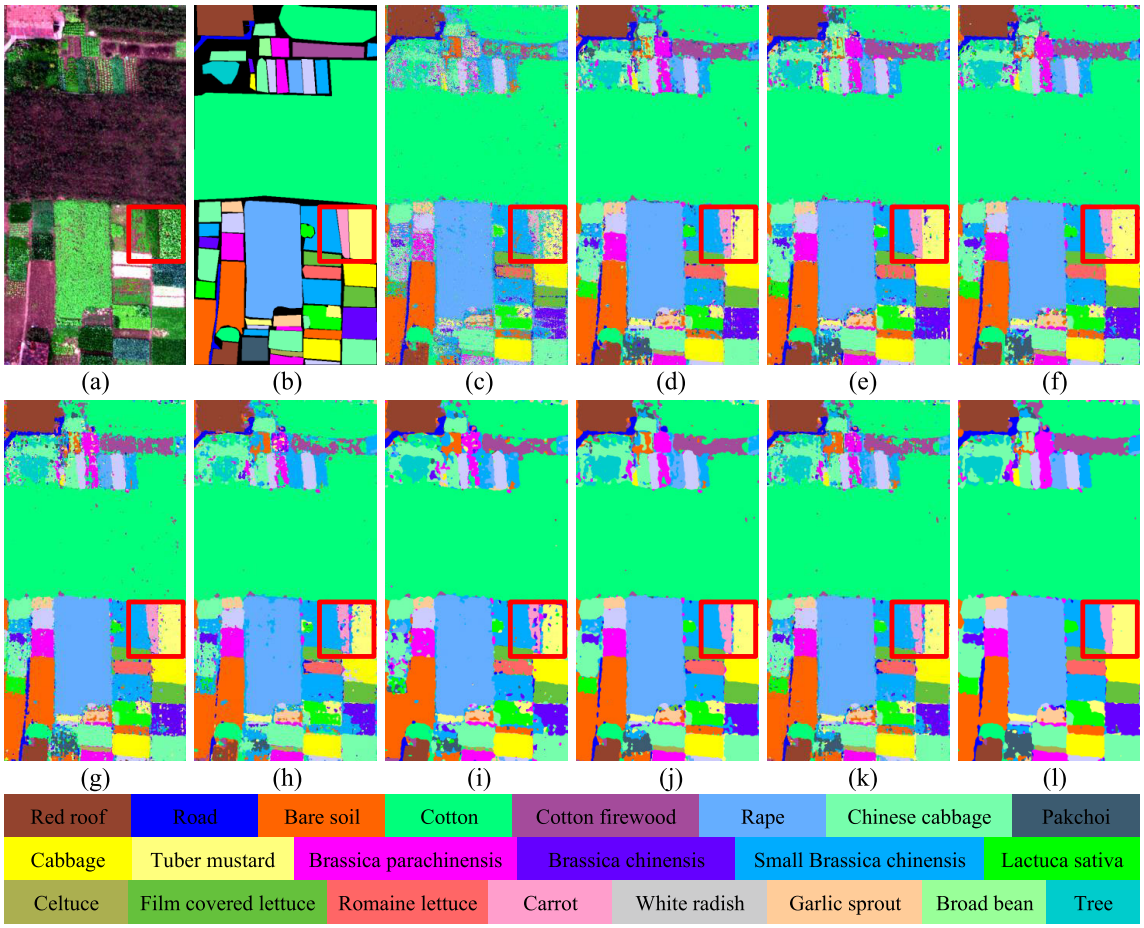



Fig. 7. Classification maps of different methods for the WHU-Hi-HongHu dataset. (a) False-color image. (b) Ground truth. (c) SVM (OA = 84.02%). (d) 2-D CNN (OA = 94.48%). (e) 3-D-CNN (OA = 94.94%). (f) SSRN (OA = 96.08%). (g) SSFTT (OA = 94.57%). (h) $S^2$GFormer (OA = 90.26%). (i) WFCG (OA = 95.90%). (j) CW-SAGE (OA = 91.68%). (k) miniGCN (OA = 93.24%). (l) NAGIN (OA = 98.34%).

TABLE VI

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, $\mathcal{K}$, AND THE ACCURACIES OF EACH CLASS FOR THE WHU-HI-HONGHU DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

| Class | SVM | 2D-CNN | 3D-CNN | SSRN | SSFTT | $S^2$GFormer | WFCG | CW-SAGE | miniGCN | NAGIN |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 91.75 | 96.18 | 97.43 | 97.59 | 96.93 | 88.37 | 97.43 | 94.72 | 96.41 | **98.14** |
| 2 | 81.38 | 80.81 | **88.75** | 87.16 | 84.25 | 61.77 | 79.83 | 55.29 | 75.60 | 87.08 |
| 3 | 90.03 | 94.39 | 95.97 | 95.96 | 95.68 | 87.70 | 97.72 | 93.73 | 95.09 | **98.56** |
| 4 | 98.74 | 99.44 | 99.54 | 99.67 | 99.35 | **99.83** | 99.78 | 99.75 | 99.58 | 99.79 |
| 5 | 36.80 | 83.08 | 85.00 | 87.76 | 85.99 | 58.65 | 85.82 | 76.77 | 75.74 | **94.38** |
| 6 | 90.67 | 97.03 | 97.63 | 98.72 | 98.17 | 88.38 | 98.02 | 94.20 | 96.19 | **99.73** |
| 7 | 76.00 | 92.75 | 93.37 | 93.93 | 90.98 | 92.45 | 95.92 | 89.61 | 92.42 | **98.68** |
| 8 | 11.37 | 68.11 | 70.25 | 74.59 | 58.34 | 25.42 | 63.62 | 36.71 | 49.41 | **90.80** |
| 9 | 92.68 | 98.09 | 97.67 | 98.71 | 96.89 | 89.18 | 98.70 | 96.25 | 98.22 | **99.87** |
| 10 | 55.16 | 89.53 | 86.27 | 92.50 | 91.46 | 82.51 | 91.24 | 78.58 | 86.55 | **97.07** |
| 11 | 42.53 | 83.04 | 88.86 | 89.17 | 83.52 | 77.92 | 91.76 | 81.59 | 82.88 | **96.05** |
| 12 | 55.07 | 81.46 | 81.33 | 83.33 | 79.45 | 67.59 | 80.93 | 64.32 | 68.57 | **91.87** |
| 13 | 69.35 | 85.90 | 85.85 | 90.30 | 86.81 | 91.98 | 87.83 | 85.44 | 81.64 | **95.33** |
| 14 | 61.61 | 91.86 | 87.48 | 93.56 | 89.33 | 63.29 | 92.57 | 77.03 | 87.19 | **96.84** |
| 15 | 27.54 | 81.14 | 81.44 | 88.32 | 85.93 | 71.14 | 93.84 | 55.90 | 91.12 | **95.86** |
| 16 | 87.66 | 97.96 | 96.31 | **98.28** | 95.41 | 94.58 | 96.62 | 94.94 | 95.20 | 98.09 |
| 17 | 58.77 | 92.36 | 86.64 | 95.91 | 94.25 | 88.75 | 95.47 | 91.21 | 91.17 | **99.03** |
| 18 | 42.56 | 89.80 | 94.37 | 92.23 | 91.54 | 80.21 | 94.54 | 68.75 | 92.53 | **96.86** |
| 19 | 76.50 | 90.35 | 90.30 | 91.98 | 93.70 | 90.57 | 94.40 | 89.22 | 90.83 | **95.69** |
| 20 | 59.64 | 82.01 | 83.19 | 85.31 | 78.28 | 82.90 | 85.54 | 75.46 | 77.72 | **96.29** |
| 21 | 2.41 | 61.30 | 64.91 | 72.36 | 41.49 | 0.00 | 45.36 | 8.07 | 11.57 | **88.05** |
| 22 | 40.87 | 77.85 | 87.72 | 87.20 | 83.44 | 61.44 | 93.67 | 84.05 | 84.27 | **97.55** |
| OA(%) | 84.02 | 94.48 | 94.84 | 96.08 | 94.57 | 90.26 | 95.90 | 91.68 | 93.24 | **98.34** |
| AA(%) | 61.32 | 87.02 | 88.20 | 90.66 | 86.42 | 74.76 | 89.12 | 76.89 | 82.72 | **95.98** |
| $\mathcal{K} \times 100$ | 79.56 | 93.02 | 93.60 | 95.04 | 93.11 | 87.60 | 94.80 | 89.44 | 91.42 | **97.90** |

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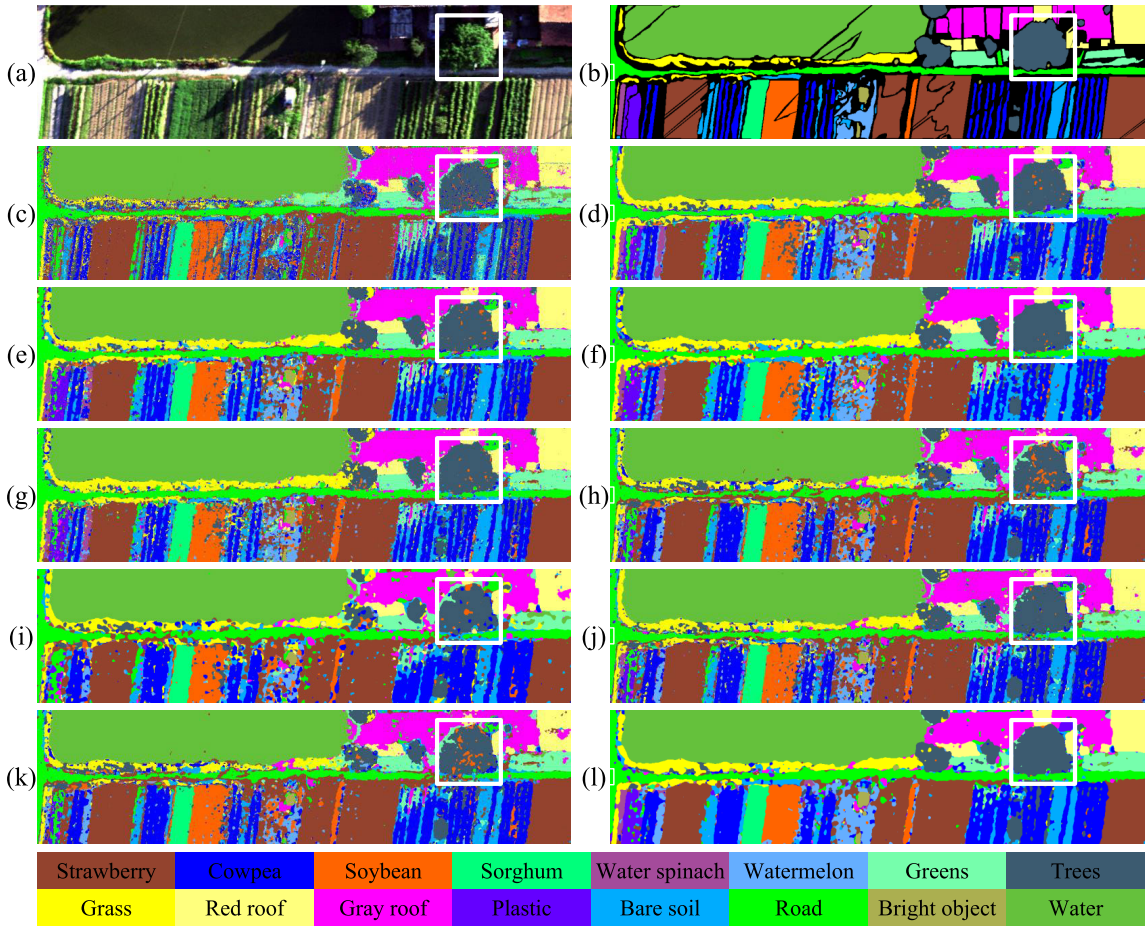



Fig. 8. Classification maps of different methods for the WHU-Hi-HanChuan dataset. (a) False-color image. (b) Ground truth. (c) SVM (OA = 82.48%). (d) 2-D CNN (OA = 93.32%). (e) 3-D-CNN (OA = 94.04%). (f) SSRN (OA = 94.84%). (g) SSFTT (OA = 93.11%). (h) $S^2$GFormer (OA = 88.49%). (i) WFCG (OA = 92.12%). (j) CW-SAGE (OA = 86.98%). (k) miniGCN (OA = 92.54%). (l) NAGIN(OA = 96.29%).

of WFCG and miniGCN. In comparison, the NAGIN model of this article not only achieves the highest test accuracy but also achieves the best classification results at the boundary and inside of the land cover. As shown in the red outlined area in Fig. 7. Within the red outlined area, comparative algorithms exhibit varying degrees of noise, whereas NAGIN shows the least noise.

*3) Results on WHU-Hi-HanChuan:* The classification accuracy of different methods on the WHU-Hi-HanChuan dataset is listed in Table VII. It can be seen that the OA, AA, and $\mathcal{K}$ of the proposed method are higher than the nine comparison methods. Among the 16 land-cover classes, the classification accuracy of 12 land-cover classes exceeds that of other methods. Only the classification accuracies of the four land-cover classes, greens, red roof, gray roof, and bare soil, are slightly different from the best classification accuracy of the comparison method. The OA of SVM is 82.48%. The best of the three CNN-based classification methods is SSRN, with an OA of 94.84%. The best of the four GCN-based classification methods is miniGCN, with an OA of 92.54%. The OA of SSFTT is 93.11%, and the OA of the NAGIN classification model we proposed is 96.29%.

The classification results of different methods are shown in Fig. 8. For the convenience of comparison, we show the false-color image and ground-truth map of the WHU-Hi-HanChuan dataset. It can be seen that the classification results of SVM are the worst, and obvious salt-and-pepper noise can be seen. Among the three CNN-based classification methods, SSRN achieves the best performance. The performance of SSFTT shows little difference compared to 2-D CNN and 3-D CNN. Of the four GCN-based comparative methods, CW-SAGE delivers the best classification outcome. Among the comparative methods, SSRN achieves the best classification performance, but its results are still inferior to NAGIN in most categories. NAGIN demonstrates the best classification accuracy across all classes. In the white outlined area in Fig. 8, there is a patch of trees where the comparative methods exhibit varying degrees of internal noise, whereas NAGIN shows almost no noise.

### *D. NAB Effectiveness Analysis*

NAB optimizes the adjacency matrix and improves classification performance by removing anomalous pixels from neighboring nodes. To evaluate the effectiveness of NAB, we conducted experimental verification on three datasets (Indian Pines, WHU-Hi-HongHu, and WHU-Hi-HanChuan), selecting ten training samples for each category. The proposed NAB method was compared with four normalization

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TABLE VII

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, $\mathcal{K}$, AND THE ACCURACIES OF EACH CLASS FOR THE WHU-HI-HANCHUAN DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

| Class | SVM | 2D-CNN | 3D-CNN | SSRN | SSFTT | S$^2$GFormer | WFCG | CW-SAGE | miniGCN | NAGIN |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 95.04 | 97.06 | 97.07 | 97.56 | 97.40 | 98.45 | 97.87 | 95.99 | 97.71 | **98.58** |
| 2 | 71.38 | 89.90 | 91.92 | 86.62 | 89.32 | 90.21 | 90.57 | 88.29 | 92.64 | **95.13** |
| 3 | 72.71 | 85.11 | 92.57 | 94.00 | 84.21 | 92.15 | 87.66 | 81.14 | 88.71 | **94.61** |
| 4 | 91.71 | 96.26 | 96.11 | 97.81 | 97.20 | 93.58 | 97.04 | 95.45 | 96.23 | **98.28** |
| 5 | 2.67 | 70.33 | 76.00 | 77.83 | 79.42 | 11.36 | 47.22 | 22.81 | 55.98 | **85.27** |
| 6 | 10.41 | 74.19 | 62.34 | 74.17 | 60.64 | 46.47 | 59.01 | 37.06 | 70.02 | **86.34** |
| 7 | 82.72 | 90.01 | 89.14 | **90.02** | 88.04 | 87.97 | 86.05 | 79.91 | 83.52 | 86.12 |
| 8 | 68.08 | 91.23 | 90.71 | 93.91 | 90.20 | 87.01 | 90.37 | 76.23 | 89.97 | **95.61** |
| 9 | 40.10 | 89.33 | 89.94 | 89.13 | 89.39 | 59.46 | 80.50 | 62.81 | 88.93 | **91.91** |
| 10 | 89.13 | 98.67 | 98.10 | **98.81** | 97.43 | 81.78 | 97.01 | 85.94 | 97.19 | 98.59 |
| 11 | 90.82 | 96.76 | 97.13 | **98.63** | 96.71 | 92.66 | 95.91 | 88.05 | 93.36 | 97.87 |
| 12 | 11.50 | 63.31 | 77.93 | 81.63 | 65.15 | 28.17 | 43.57 | 17.98 | 46.13 | **83.31** |
| 13 | 47.62 | 72.95 | 73.53 | **82.57** | 72.86 | 62.80 | 66.52 | 55.91 | 64.41 | 81.72 |
| 14 | 83.10 | 90.02 | 91.84 | 93.89 | 91.43 | 81.01 | 89.33 | 86.95 | 89.56 | **95.67** |
| 15 | 29.23 | 60.48 | 72.27 | 71.21 | 73.50 | 50.18 | 65.57 | 48.13 | 57.30 | **81.67** |
| 16 | 98.92 | 99.53 | 99.72 | 99.75 | 99.47 | 98.98 | 99.52 | 99.24 | 99.50 | **99.76** |
| OA(%) | 82.48 | 93.32 | 94.04 | 94.84 | 93.11 | 88.49 | 92.12 | 86.98 | 92.54 | **96.29** |
| AA(%) | 61.58 | 85.32 | 87.27 | 89.22 | 85.77 | 72.25 | 80.86 | 70.12 | 81.95 | **91.90** |
| $\mathcal{K} \times 100$ | 79.39 | 92.18 | 93.02 | 93.96 | 91.93 | 86.47 | 90.75 | 84.70 | 91.25 | **95.66** |

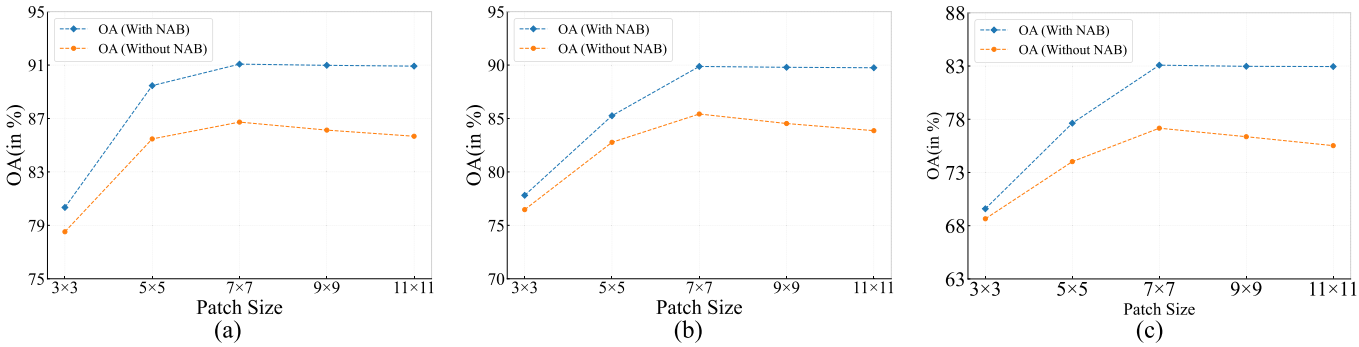



Fig. 9. OA with and without NAB at different patch sizes. (a) Indian Pines. (b) WHU-Hi-HongHu. (c) WHU-Hi-HanChuan.

approaches (normalize, maxabs, minmax, and robust). Normalize standardizes samples by dividing them by their $L2$ norm, maxabs scales features using their maximum absolute values, minmax adjusts the data range through linear transformation, and robust scaling is based on median and interquartile range. NAB achieves standardization by subtracting the mean and dividing by the standard deviation. In addition, the classification performance without NAB was also evaluated, as shown in Table VIII. It can be seen that the classification accuracy of anomalous pixels in neighboring nodes is the lowest of the three datasets. After using different methods to remove anomalous pixels in neighboring nodes, the classification accuracy has been improved to varying degrees. Normalize improves classification accuracy the least, while scale improves it the most. In this article, NAB uses the scale method to remove anomalous pixels in neighboring nodes. After using NAB, Indian Pines dataset OA improves by 4.34%, AA improves by 2.75%, $\mathcal{K}$ improves by 4.16%, WHU-Hi-HongHu dataset OA improves by 4.45%, AA improves by 4.23%, $\mathcal{K}$ improves by 5.4%, WHU-Hi-HanChuan dataset OA improves by 5.92%, AA improves by 7.9%, and $\mathcal{K}$ improves by 7.02%.

To further verify the effectiveness of NAB, OA with and without NAB was tested under different patch sizes, as shown in Fig. 9. HSI classification models typically employ image patches as input. Larger patch sizes contain richer information, which generally enhances classification performance. However, excessively large patches may incorporate anomalous pixels that degrade classification effectiveness. As shown by the orange dashed lines in Fig. 9, the classification accuracy on all three datasets initially increases but then decreases with growing patch size. The proposed NAB effectively addresses this issue by removing anomalous pixels from neighboring nodes. The blue dashed lines in Fig. 9 demonstrate that incorporating NAB significantly improves classification accuracy across all three datasets. Notably, the classification accuracy maintains stable performance without evident degradation as patch size increases, proving NAB's capability to effectively eliminate anomalous pixels from neighboring nodes and consequently enhance classification performance.

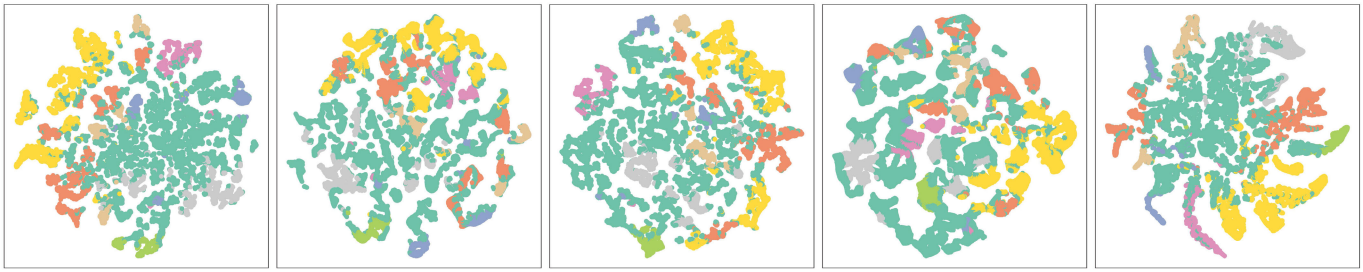
### *E. GIN Effectiveness Analysis*

To validate the effectiveness of GIN in node aggregation, we compared it with four classic GCN aggregation approaches, including S$^2$GFormer, WFCG, CW-SAGE, and miniGCN. S$^2$GFormer adapts the transformer architecture to graph-structured data, combining global attention and positional encoding to capture long-range dependencies between nodes. WFCG dynamically learns the importance weights

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TABLE VIII
ABLATION STUDY ON THE ACCURACY METRICS OF NAB

| Methods | Indian Pines(10/per class) | | | WHU-Hi-HongHu(10/per class) | | | WHU-Hi-HanChuan(10/per class) | | |
|---|---|---|---|---|---|---|---|---|---|
| | OA(%) | AA(%) | $\mathcal{K} \times 100$ | OA(%) | AA(%) | $\mathcal{K} \times 100$ | OA(%) | AA(%) | $\mathcal{K} \times 100$ |
| w/o NAB | 86.73 | 91.99 | 84.96 | 85.42 | 84.77 | 81.91 | 77.16 | 72.27 | 73.31 |
| Normalize | 87.32 | 92.42 | 85.64 | 86.17 | 84.02 | 82.68 | 78.51 | 75.23 | 75.11 |
| Maxabs | 87.97 | 92.57 | 86.35 | 86.96 | 86.03 | 83.74 | 79.75 | 76.68 | 76.53 |
| Minmax | 88.40 | 92.84 | 86.85 | 87.69 | 85.42 | 84.64 | 80.77 | 77.70 | 77.69 |
| Robust | 89.51 | 93.98 | 88.09 | 88.42 | 87.13 | 85.51 | 81.24 | 77.92 | 78.23 |
| **NAB** | **91.07** | **94.74** | **89.12** | **89.87** | **89.00** | **87.31** | **83.08** | **80.17** | **80.33** |



(a) (b) (c) (d) (e)

Fig. 10. t-SNE visualizations of feature representations aggregated by different methods. (a) $S^2$GFormer. (b) WFCG. (c) CW-SAGE. (d) miniGCN. (e) GIN.

TABLE IX
ABLATION STUDY ON THE ACCURACY METRICS OF GIN

| Methods | Indian Pines | | | WHU-Hi-HongHu | | | WHU-Hi-HanChuan | | |
|---|---|---|---|---|---|---|---|---|---|
| | OA(%) | AA(%) | $\mathcal{K} \times 100$ | OA(%) | AA(%) | $\mathcal{K} \times 100$ | OA(%) | AA(%) | $\mathcal{K} \times 100$ |
| w/o GIN | 84.72 | 82.64 | 83.33 | 89.36 | 80.71 | 87.52 | 87.18 | 72.46 | 85.3 |
| S2GFormer | 85.13 | 84.50 | 82.89 | 90.26 | 74.76 | 87.60 | 88.49 | 72.25 | 86.47 |
| WFCG | 88.76 | 85.91 | 87.09 | 95.90 | 89.12 | 94.80 | 92.12 | 80.86 | 90.75 |
| CW-SAGE | 88.25 | 83.85 | 86.49 | 91.68 | 76.89 | 89.44 | 86.98 | 70.12 | 84.70 |
| miniGCN | 88.85 | 87.80 | 87.21 | 93.24 | 82.72 | 91.42 | 92.54 | 81.95 | 91.25 |
| **GIN** | **91.07** | **89.10** | **89.82** | **98.34** | **95.98** | **97.90** | **96.29** | **91.90** | **95.66** |

of neighboring nodes through an attention mechanism. CW-SAGE reduces computational complexity and improves model generalization by sampling a fixed number of neighboring nodes. The approach adopted by miniGCN is grounded in spectral graph theory and aggregates information from neighboring nodes using a normalized weighted average strategy. In contrast, GIN is based on the WL graph isomorphism test theory and obtains multiple representations of nodes through summation aggregation. Meanwhile, we compared the classification accuracy without using GIN. As shown in Table IX, it can be observed that the classification performance of GIN is significantly better than several comparative algorithms.

The features aggregated by different methods are shown in Fig. 10. It can be seen that the features extracted by $S^2$GFormer, WFCG, CW-SAGE, and miniGCN exhibit relatively low discriminability, with a significant degree of overlap between their feature distributions. In contrast, the features extracted by GIN show clear discriminability. This indicates that although $S^2$GFormer and WFCG are capable of learning the weights of neighboring nodes at different positions, in hyperspectral classification tasks, the weights of neighboring nodes are not strongly correlated with their positions. Therefore, attention-based aggregation methods do not lead to more refined node features in this context. CW-SAGE and miniGCN essentially use average aggregation to combine features of neighboring nodes, which fails to capture the heterogeneity among them. On the other hand, GIN integrates the advantages of both average aggregation and attention mechanisms, considering the importance of neighboring nodes while ignoring their positional information, making it more suitable for hyperspectral node classification tasks.

Furthermore, we analyzed the effect of different numbers of graph convolutional layers on classification accuracy on three datasets (Indian Pines, WHU-Hi-HongHu, and WHU-Hi-HanChuan). All three datasets take 1% training samples, and their classification accuracy is shown in Fig. 11. It can be seen that the effect of one layer of graph convolution is poor. The OA, AA, and $\mathcal{K}$ of Indian Pines are 61.26%, 52.41%, and 54.74%, respectively. The OA, AA, and $\mathcal{K}$ of WHU-Hi-HongHu are 86.38%, 61.85%, and 82.69%, respectively. The OA, AA, and $\mathcal{K}$ of WHU-Hi-HanChuan are 77.56%, 58.76%, and 73.31%, respectively. After using two layers of graph convolution, the classification accuracy is significantly improved. The OA, AA, and $\mathcal{K}$ of Indian Pines are all above 89%, the OA, AA, and $\mathcal{K}$ of WHU-Hi-HongHu are all above 95%, and the OA, AA, and $\mathcal{K}$ of WHU-Hi-HanChuan are all above 91%. After using three layers of graph convolution, the classification accuracy of the model no longer improves on the three datasets. The GIN enhances the representational capacity of nodes by leveraging graph

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TABLE X

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, $\mathcal{K}$, AND THE ACCURACIES OF EACH CLASS FOR THE HOUSTON 2018 DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

| Class | SVM | 2D-CNN | 3D-CNN | SSRN | SSFTT | $S^2$GFormer | WFCG | CW-SAGE | miniGCN | NAGIN |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 80.34 | 83.30 | 79.31 | 73.05 | 63.12 | 86.00 | 84.58 | **89.09** | 87.24 | 79.32 |
| 2 | **93.21** | 90.79 | 89.37 | 92.26 | 91.78 | 91.37 | 89.83 | 92.54 | 91.51 | 89.33 |
| 3 | **100.0** | 94.15 | 96.35 | 88.01 | 87.43 | **100.0** | 99.27 | **100.0** | 99.85 | **100.0** |
| 4 | 91.14 | **98.17** | 97.20 | 96.93 | 94.13 | 94.18 | 92.79 | 92.08 | 91.65 | 90.24 |
| 5 | 20.04 | 79.69 | 79.65 | 67.28 | 78.93 | 89.79 | 82.69 | **94.68** | 92.64 | 88.65 |
| 6 | 52.68 | 97.83 | 98.34 | 99.34 | 92.18 | 99.61 | 91.68 | **99.95** | 99.77 | 97.70 |
| 7 | 66.17 | 95.49 | 95.86 | **100.0** | 50.00 | **100.0** | 90.56 | 99.25 | 97.04 | 94.83 |
| 8 | 81.51 | 88.85 | 90.58 | **96.73** | 89.72 | 86.36 | 90.33 | 90.02 | 88.23 | 95.24 |
| 9 | 92.34 | 97.80 | 97.42 | 92.48 | **97.83** | 93.75 | 95.53 | 94.68 | 94.14 | 97.58 |
| 10 | 31.97 | 75.83 | 75.50 | 57.62 | 72.15 | 73.93 | 72.06 | 77.16 | 76.33 | **84.32** |
| 11 | 43.98 | 66.66 | 65.66 | 54.07 | 68.67 | 69.05 | 66.92 | 68.00 | 68.64 | **78.93** |
| 12 | 0.00 | 11.07 | 14.43 | 7.97 | 18.91 | 0.00 | 19.68 | 29.58 | 17.39 | **37.10** |
| 13 | 62.84 | 81.64 | 84.50 | **91.71** | 82.03 | 72.88 | 75.60 | 75.86 | 72.51 | 87.02 |
| 14 | 41.69 | 87.81 | 88.55 | 80.85 | 87.13 | 90.75 | 79.00 | 88.07 | 84.82 | **94.65** |
| 15 | 0.00 | 96.42 | 98.08 | 98.88 | 96.18 | 98.60 | 98.35 | **99.50** | 99.32 | 97.58 |
| 16 | 0.00 | 90.43 | 88.79 | 85.32 | 88.23 | 93.54 | 85.14 | 94.37 | 92.00 | **94.71** |
| 17 | 0.00 | 34.93 | 92.47 | 35.62 | 78.08 | 0.00 | 82.67 | **100.0** | **100.0** | 95.83 |
| 18 | 0.05 | 82.92 | 83.56 | 88.28 | 79.00 | **93.64** | 84.70 | 92.27 | 91.59 | 87.65 |
| 19 | 7.17 | 88.06 | 89.53 | 77.30 | 88.60 | 98.34 | 90.14 | **99.65** | 97.09 | 99.37 |
| 20 | 85.16 | 98.89 | 98.64 | 99.19 | 95.38 | 99.27 | 98.04 | **99.51** | 99.14 | 99.09 |
| OA(%) | 71.88 | 89.67 | 89.66 | 85.83 | 88.93 | 87.77 | 87.83 | 89.11 | 88.13 | **92.65** |
| AA(%) | 47.51 | 82.04 | 85.19 | 79.14 | 79.97 | 81.55 | 83.48 | 88.81 | 87.04 | **89.46** |
| $\mathcal{K} \times 100$ | 61.83 | 86.51 | 86.51 | 81.76 | 85.52 | 83.94 | 84.13 | 85.72 | 84.43 | **90.43** |

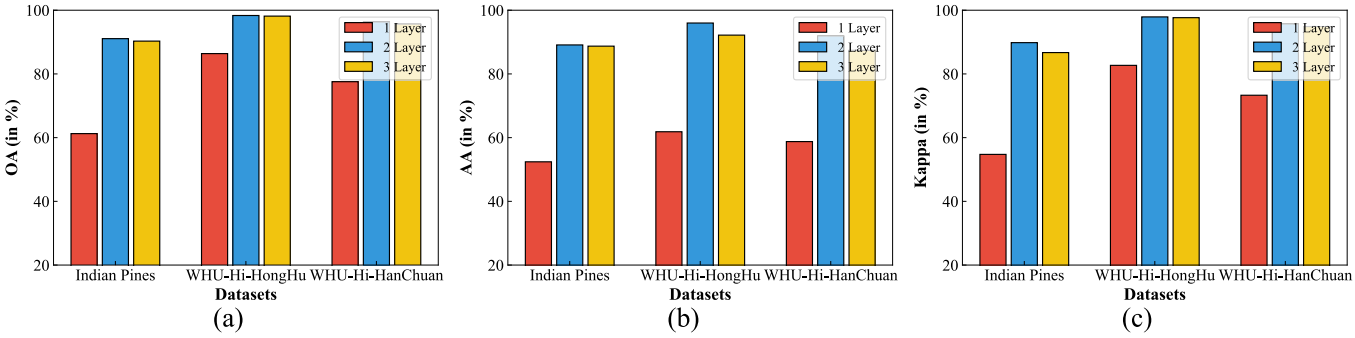



Fig. 11. Classification accuracies of different numbers of graph convolutional layers. (a) OA of the three datasets. (b) AA of the three datasets. (c) Kappa of the three datasets.

isomorphism, effectively capturing local structural and spectral differences, which significantly improves the classification accuracy of HSI. However, as the number of network layers increases, node features tend to become over-smoothed and homogenized, thereby weakening the discriminative power between different classes and degrading overall classification performance. Therefore, this article employs a two-layer graph convolutional structure to extract node features, balancing expressive power, and feature diversity.

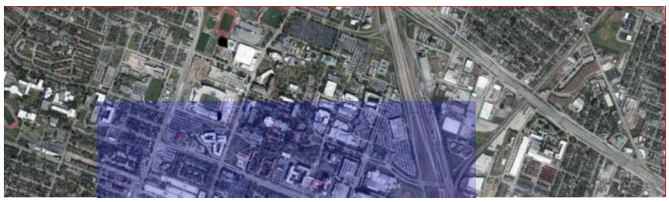


Fig. 12. Labeled regions in the Houston 2018 dataset.

### *F. Generalization Performance Analysis*

Typically, the HSI classification experiment selects training and testing samples in the same scene, so the model can learn the ground features of the scene relatively easily, but the classification performance of the model in other scenes needs to be further verified. To more strictly test the effectiveness of the method in this article, we train and test according to the competition rules of the 2018 IEEE GRSS data fusion competition. The model is trained in the blue shaded area of Fig. 12 and tested outside the blue shaded area, the test area is unlabeled. During the competition, the classification accuracy is obtained from the official website by uploading the classification maps. In this article, we evaluate the classification performance of each model in the testing area through visual results, as shown in Fig. 13. In addition, we also provide the accuracy of selecting 1% of the samples for training in the training area, as shown in Table X.

The worst classification result in Fig. 13 is $S^2$GFormer, while the SVM with the lowest classification accuracy in Table X. Further analysis reveals that SVM maintains relatively stable classification performance, with consistent results

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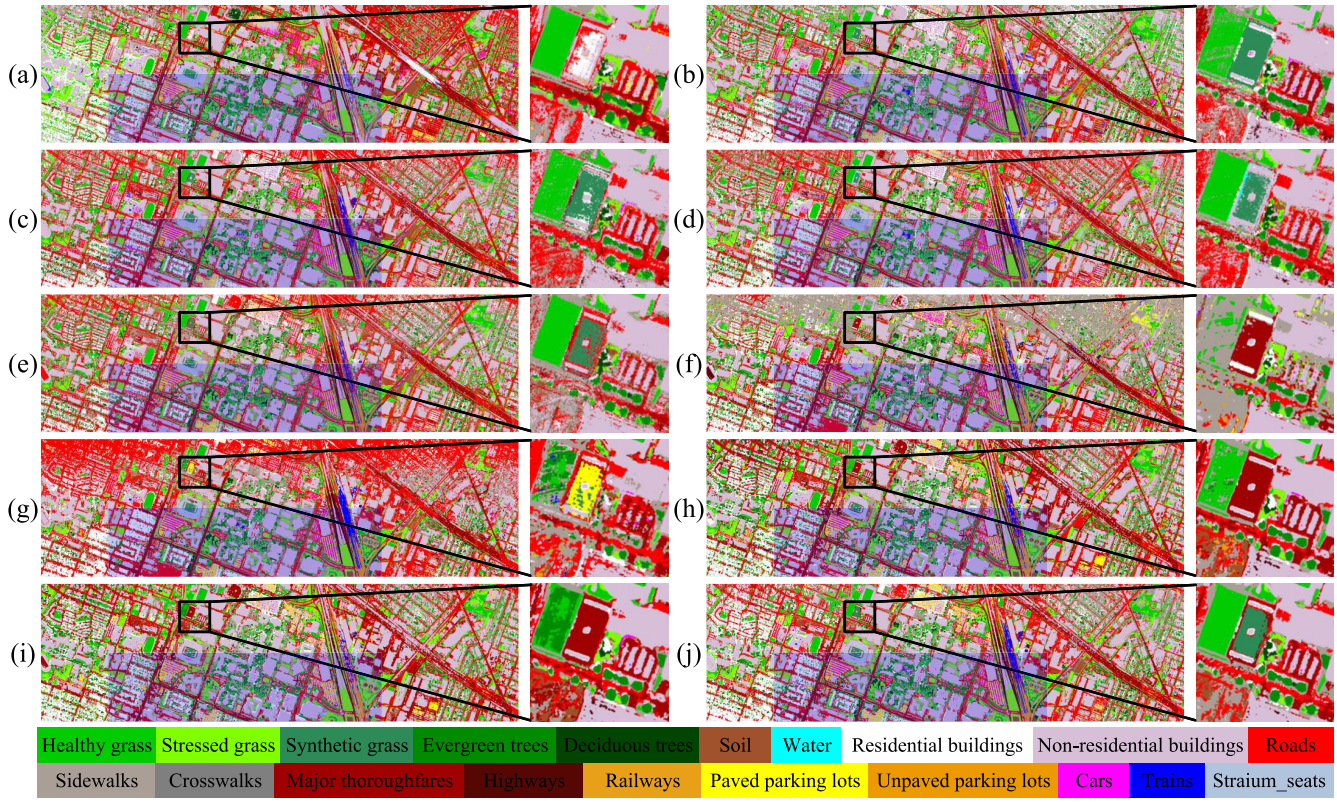



Fig. 13. Classification maps of different methods for the Houston 2018 dataset. (a) SVM (OA = 80.64%). (b) 2-DCNN (OA = 92.04%). (c) 3D-CNN (OA = 89.98%). (d) SSRN (OA = 82.50%). (e) SSFTT (OA = 89.49%). (f) S2GFormer (OA = 88.11%). (g) WFCG (OA = 89.66%). (h) CW-SAGE (OA = 91.48%). (i) miniGCN (OA = 90.05%). (j) NAGIN (OA = 93.60%).

between training and test regions. While the classification accuracy of $S^2$GFormer is higher in the training area, the classification results in the test area are obviously worse than those in the training area, and there is a serious overfitting phenomenon. The classification accuracy of the three CNN-based classification methods is not much different in the training area, but there is still a significant difference in the test area, as shown in the enlarged area in Fig. 13. Among the four GCN-based comparison methods, WFCG also has a serious overfitting phenomenon. The overall classification results of CW-SAGE and miniGCN are not much different. CW-SAGE has a better classification result because it uses a random sampling strategy. In comparison, the NAGIN proposed in this article not only achieves the highest classification accuracy in the training area but also achieves the best classification results in the testing area.

## V. Conclusion

We propose an NAGIN framework for HSI classification, which can effectively construct a diversified feature representation of land cover. To mitigate the negative impact of anomalous pixels on classification accuracy, NAB is employed to optimize the adjacency matrix. NAB can more accurately remove anomalous pixels in neighboring nodes by reconstructing spectral features. To enhance node feature extraction, a graph aggregation strategy tailored to HSI is developed based on graph isomorphism theory, allowing for the capture of diverse feature representations. Furthermore, KAN is introduced to improve the model's feature extraction capability. Extensive experiments demonstrate that NAGIN is effective in optimizing graph structures. Its node aggregation strategy significantly improves HSI classification accuracy and exhibits strong generalization ability. Future research directions include refining graph node aggregation mechanisms and developing more sophisticated spectral feature extraction models to enable finer-grained HSI classification.

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